3.2 Important Discrete Distributions

Several discrete probability distributions arise frequently in statistical quality control. In this section, we discuss the hypergeometric distribution, the binomial distribution, the Poisson distribution, and the Pascal or negative binomial distribution.

3.2.1 The Hypergeometric Distribution

Suppose that there is a finite population consisting of N items. Some number—say, $D (D \leq N)$—of these items fall into a class of interest. A random sample of n items is selected from the population *without replacement*, and the number of items in the sample that fall into the class of interest—say, x—is observed. Then x is a hypergeometric random variable with the probability distribution defined as follows.

Definition

The **hypergeometric probability distribution** is

$$p(x) = \frac{\binom{D}{x}\binom{N-D}{n-x}}{\binom{N}{n}} \qquad x = 0,1,2,\ldots,\min(n,D) \tag{3.8}$$

The mean and variance of the distribution are

$$\mu = \frac{nD}{N} \tag{3.9}$$

and

$$\sigma^2 = \frac{nD}{N}\left(1-\frac{D}{N}\right)\left(\frac{N-n}{N-1}\right) \tag{3.10}$$

In the above definition, the quantity

$$\binom{a}{b} = \frac{a!}{b!(a-b)!}$$

is the number of combinations of a items taken b at a time.

The hypergeometric distribution is the appropriate probability model for selecting a random sample of n items without replacement from a lot of N items of which D are nonconforming or defective. By a random sample, we mean a sample that has been selected in such a way that all possible samples have an equal chance of being chosen. In these applications, x usually represents the number of nonconforming items found in the sample. For example, suppose that a lot contains 100 items, 5 of which do not conform to requirements. If 10 items are selected at random without replacement, then the probability of finding one or fewer nonconforming items in the sample is

$$P\{x \le 1\} = P\{x = 0\} + P\{x = 1\}$$
$$= \frac{\binom{5}{0}\binom{95}{10}}{\binom{100}{10}} + \frac{\binom{5}{1}\binom{95}{9}}{\binom{100}{10}} = 0.92314$$

In Chapter 15 we show how probability models such as this can be used to design acceptance-sampling procedures.

Some computer programs can perform these calculations. The display below is the output from Minitab for calculating cumulative hypergeometric probabilities with $N = 100$, $D = 5$ (note that Minitab uses the symbol M instead of D, and $n = 5$). Minitab will also calculate the individual probabilities for each value of x.

Cumulative Distribution Function

```
Hypergeometric with N = 100, M = 5, and n = 10

 x   P(X <= x)          x   P(X <= x)
 0     0.58375          6     1.00000
 1     0.92314          7     1.00000
 2     0.99336          8     1.00000
 3     0.99975          9     1.00000
 4     1.00000         10     1.00000
 5     1.00000
```

3.2.2 The Binomial Distribution

Consider a process that consists of a sequence of n independent trials. By independent trials, we mean that the outcome of each trial does not depend in any way on the outcome of previous trials. When the outcome of each trial is either a "success" or a "failure," the trials are called **Bernoulli trials.** If the probability of "success" on any trial—say, p—is constant, then the number of "successes" x in n Bernoulli trials has the **binomial distribution** with parameters n and p, defined as follows.

Definition

The **binomial distribution** with parameters $n \ge 0$ and $0 < p < 1$ is

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x} \qquad x = 0, 1, \ldots, n \tag{3.11}$$

The mean and variance of the binomial distribution are

$$\mu = np \tag{3.12}$$

and

$$\sigma^2 = np(1-p) \tag{3.13}$$

The binomial distribution is used frequently in quality engineering. It is the appropriate probability model for sampling from an infinitely large population, where p represents the fraction of defective or nonconforming items in the population. In these applications, x usually represents the number of nonconforming items found in a random sample of n items. For example, if $p = 0.10$ and $n = 15$, then the probability of obtaining x nonconforming items is computed from equation 3.11 as follows:

Probability Density Function

Binomial with n = 15 and p = 0.1

x	P(X = x)	x	P(X = x)
0	0.205891	6	0.001939
1	0.343152	7	0.000277
2	0.266896	8	0.000031
3	0.128505	9	0.000003
4	0.042835	10	0.000000
5	0.010471		

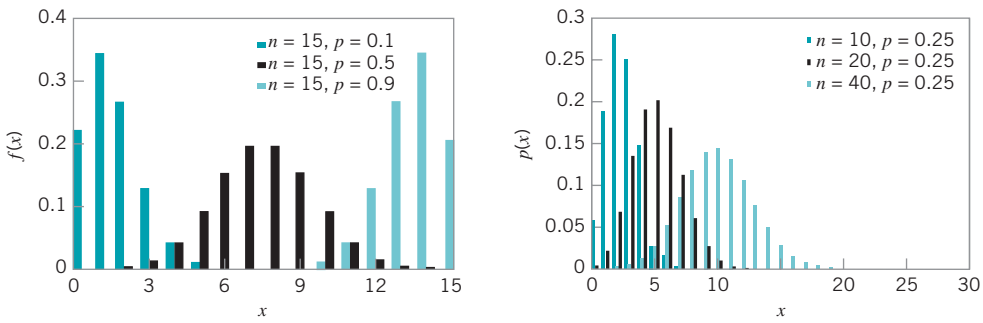
Minitab was used to perform these calculations. Notice that for all values of x that lie between $10 \leq x \leq 15$ the probability of finding x "successes" in 15 trials is zero.

Several binomial distributions are shown graphically in Fig. 3.14. The shape of those examples is typical of all binomial distributions. For a fixed n, the distribution becomes more symmetric as p increases from 0 to 0.5 or decreases from 1 to 0.5. For a fixed p, the distribution becomes more symmetric as n increases.

A random variable that arises frequently in statistical quality control is

$$\hat{p} = \frac{x}{n} \tag{3.14}$$

where x has a binomial distribution with parameters n and p. Often $\hat{p}$ is the ratio of the observed number of defective or nonconforming items in a sample (x) to the sample size (n)



(*a*) Binomial distributions for different values of p with $n = 15$.

(*b*) Binomial distributions for different values of n with $p = 0.25$.

■ **FIGURE 3.14** Binomial distributions for selected values of n and p.

and this is usually called the **sample fraction defective** or **sample fraction nonconforming.** The " ˆ " symbol is used to indicate that $\hat{p}$ is an estimate of the true, unknown value of the binomial parameter p. The probability distribution of $\hat{p}$ is obtained from the binomial, since

$$P\{\hat{p} \le a\} = P\left\{\frac{x}{n} \le a\right\} = p\{x \le na\} = \sum_{x=0}^{[na]} \binom{n}{x} p^x (1-p)^{n-x}$$

where $[na]$ denotes the largest integer less than or equal to na. It is easy to show that the mean of $\hat{p}$ is p and that the variance of $\hat{p}$ is

$$\sigma_{\hat{p}}^2 = \frac{p(1-p)}{n}$$

3.2.3 The Poisson Distribution

A useful discrete distribution in statistical quality control is the Poisson distribution, defined as follows.

Definition

The **Poisson distribution** is

$$p(x) = \frac{e^{-\lambda}\lambda^x}{x!} \qquad x = 0, 1, \ldots \tag{3.15}$$

where the parameter $\lambda > 0$. The **mean** and **variance** of the Poisson distribution are

$$\mu = \lambda \tag{3.16}$$

and

$$\sigma^2 = \lambda \tag{3.17}$$

Note that the mean and variance of the Poisson distribution are both equal to the parameter λ.

A typical application of the Poisson distribution in quality control is as a model of the number of defects or nonconformities that occur in a unit of product. In fact, any random phenomenon that occurs on a per unit (or per unit area, per unit volume, per unit time, etc.) basis is often well approximated by the Poisson distribution. As an example, suppose that the number of wire-bonding defects per unit that occur in a semiconductor device is Poisson distributed with parameter $\lambda = 4$. Then the probability that a randomly selected semiconductor device will contain two or fewer wire-bonding defects is

$$P\{x \le 2\} = \sum_{x=0}^{2} \frac{e^{-4}4^x}{x!} = 0.018316 + 0.073263 + 0.146525 = 0.238104$$

Minitab can perform these calculations. Using the Poisson distribution with the mean = 4 results in:

Probability Density Function

```
Poisson with mean = 4

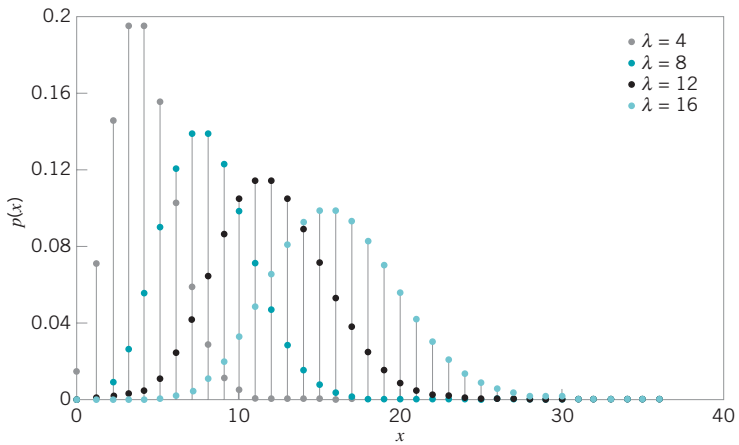
x   P(X = x)
0     0.018316
1     0.073263
2     0.146525
```

Several Poisson distributions are shown in Fig. 3.15. Note that the distribution is *skewed;* that is, it has a long tail to the right. As the parameter λ becomes larger, the Poisson distribution becomes symmetric in appearance.

It is possible to derive the Poisson distribution as a limiting form of the binomial distribution. That is, in a binomial distribution with parameters n and p, if we let n approach infinity and p approach zero in such a way that $np = \lambda$ is a constant, then the Poisson distribution results. It is also possible to derive the Poisson distribution using a pure probability argument. [For more information about the Poisson distribution, see Hines, Montgomery, Goldsman, and Borror (2004); Montgomery and Runger (2007); and the supplemental text material.]

3.2.4 The Pascal and Related Distributions

The Pascal distribution, like the binomial distribution, has its basis in Bernoulli trials. Consider a sequence of independent trials, each with probability of success p, and let x denote the trial on which the rth success occurs. Then x is a Pascal random variable with probability distribution defined as follows.



■ **FIGURE 3.15** Poisson probability distributions for selected values of λ.

Definition

The **Pascal distribution** is

$$p(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r} \qquad x = r, r+1, r+2, \ldots \tag{3.18}$$

where $r \geq 1$ is an integer. The ***mean*** and ***variance*** of the Pascal distribution are

$$\mu = \frac{r}{p} \tag{3.19}$$

and

$$\sigma^2 = \frac{r(1-p)}{p^2} \tag{3.20}$$

respectively.

Two special cases of the Pascal distribution are of interest. The first of these occurs when $r > 0$ and is not necessarily an integer. The resulting distribution is called the **negative binomial distribution.** It is relatively standard to refer to equation (3.18) as the negative binomial distribution, even when r is an integer. The negative binomial distribution, like the Poisson distribution, is sometimes useful as the underlying statistical model for various types of "count" data, such as the occurrence of nonconformities in a unit of product (see Section 7.3.1). There is an important duality between the binomial and negative binomial distributions. In the binomial distribution, we fix the sample size (number of Bernoulli trials) and observe the number of successes; in the negative binomial distribution, we fix the number of successes and observe the sample size (number of Bernoulli trials) required to achieve them. This concept is particularly important in various kinds of sampling problems.

The other special case of the Pascal distribution is if $r = 1$, in which case we have the **geometric distribution.** It is the distribution of the number of Bernoulli trials until the *first* success.

3.3 Important Continuous Distributions

In this section we discuss several continuous distributions that are important in statistical quality control. These include the normal distribution, the lognormal distribution, the exponential distribution, the gamma distribution, and the Weibull distribution.

3.3.1 The Normal Distribution

The normal distribution is probably the most important distribution in both the theory and application of statistics. If x is a normal random variable, then the probability distribution of x is defined as follows.

Definition

The **normal distribution** is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \qquad -\infty < x < \infty \tag{3.21}$$

The mean of the normal distribution is μ $(-\infty < \mu < \infty)$ and the variance is $\sigma^2 > 0$.

The normal distribution is used so much that we frequently employ a special notation, $x \sim N(\mu, \sigma^2)$, to imply that x is normally distributed with mean μ and variance σ^2. The visual appearance of the normal distribution is a symmetric, unimodal or **bell-shaped** curve and is shown in Fig. 3.16.

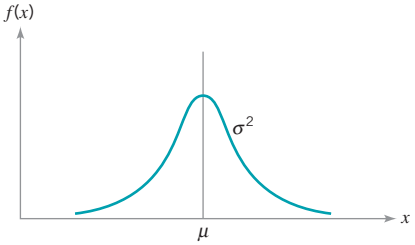
There is a simple interpretation of the standard deviation σ of a normal distribution, which is illustrated in Fig. 3.17. Note that 68.26% of the population values fall between the limits defined by the mean plus and minus one standard deviation ($\mu \pm 1\sigma$); 95.46% of the values fall between the limits defined by the mean plus and minus two standard deviations ($\mu \pm 2\sigma$); and 99.73% of the population values fall within the limits defined by the mean plus and minus three standard deviations ($\mu \pm 3\sigma$). Thus, the standard deviation measures the distance on the horizontal scale associated with the 68.26%, 95.46%, and 99.73% containment limits. It is common practice to round these percentages to 68%, 95%, and 99.7%.

The cumulative normal distribution is defined as the probability that the normal random variable x is less than or equal to some value a, or

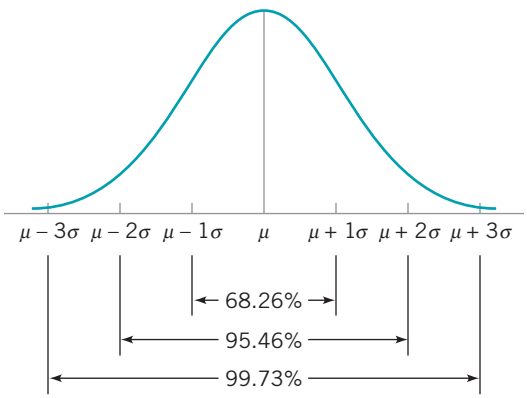
$$P\{x \le a\} = F(a) = \int_{-\infty}^{a} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \tag{3.22}$$

This integral cannot be evaluated in closed form. However, by using the change of variable

$$z = \frac{x - \mu}{\sigma} \tag{3.23}$$



■ **FIGURE 3.16** The normal distribution.



■ **FIGURE 3.17** Areas under the normal distribution.

the evaluation can be made independent of μ and σ^2. That is,

$$P\{x \le a\} = P\left\{z \le \frac{a-\mu}{\sigma}\right\} \equiv \Phi\left(\frac{a-\mu}{\sigma}\right)$$

where $\Phi(\cdot)$ is the cumulative distribution function of the **standard normal distribution** (mean = 0, standard deviation = 1). A table of the cumulative standard normal distribution is given in Appendix Table II. The transformation (3.23) is usually called **standardization,** because it converts a $N(\mu, \sigma^2)$ random variable into an $N(0, 1)$ random variable.

EXAMPLE 3.7 Tensile Strength of Paper

The tensile strength of paper used to make grocery bags is an important quality characteristic. It is known that the strength—say, x—is normally distributed with mean $\mu = 40$ lb/in^2 and standard deviation $\sigma = 2$ lb/in^2, denoted $x \sim N(40, 2^2)$. The purchaser of the bags requires them to have a strength of at least 35 lb/in^2. Calculate the probability that bags produced from this paper will meet or exceed the specification.

SOLUTION

The probability that a bag produced from this paper will meet or exceed the specification is $P\{x \ge 35\}$. Note that

$$P\{x \ge 35\} = 1 - P\{x \le 35\}$$

To evaluate this probability from the standard normal tables, we standardize the point 35 and find

$$P\{x \le 35\} = P\left\{z \le \frac{35-40}{2}\right\} =$$

$$P\{z \le -2.5\} = \Phi(-2.5) = 0.0062$$

Consequently, the desired probability is

$$P\{x \ge 35\} = 1 - P\{x \le 35\} = 1 - 0.0062 = 0.9938$$

Figure 3.18 shows the tabulated probability for both the $N(40, 2^2)$ distribution and the standard normal distribution. Note that the shaded area to the left of 35 lb/in^2 in Fig. 3.18 represents the fraction nonconforming or "fallout" produced by the bag manufacturing process.

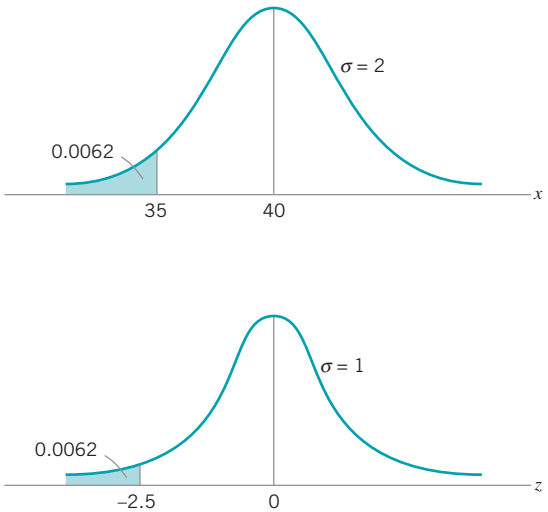


FIGURE 3.18 Calculation of $P\{x \le 35\}$ in Example 3.7.

In addition to the appendix table, many computer programs can calculate normal probabilities. Minitab has this capability.

Appendix Table II gives only probabilities to the left of positive values of z. We will need to utilize the symmetry property of the normal distribution to evaluate probabilities. Specifically, note that

$$P\{x \ge a\} = 1 - P\{x \le a\} \tag{3.24}$$

$$P\{x \le -a\} = P\{x \le a\} \tag{3.25}$$

and

$$P\{x \ge -a\} = P\{x \le a\} \tag{3.26}$$

It is helpful in problem solution to draw a graph of the distribution, as in Fig. 3.18.

EXAMPLE 3.8 Shaft Diameters

The diameter of a metal shaft used in a disk-drive unit is normally distributed with mean 0.2508 in. and standard deviation 0.0005 in. The specifications on the shaft have been established as 0.2500 ± 0.0015 in. What fraction of the shafts produced conform to specifications?

SOLUTION

The appropriate normal distribution is shown in Fig. 3.19. Note that

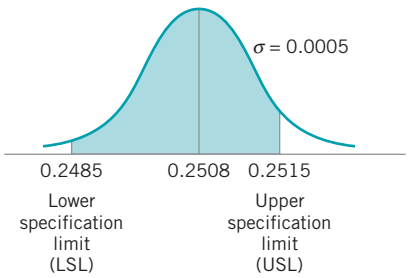
$$\begin{aligned} P\{0.2485 \le x \le 0.2515\} &= P\{x \le 0.2515\} - P\{x \le 0.2485\} \\ &= \Phi\left(\frac{0.2515 - 0.2508}{0.0005}\right) - \Phi\left(\frac{0.2485 - 0.2508}{0.0005}\right) \\ &= \Phi(1.40) - \Phi(-4.60) \\ &= 0.9192 - 0.0000 \\ &= 0.9192 \end{aligned}$$

Thus, we would expect the process yield to be approximately 91.92%; that is, about 91.92% of the shafts produced conform to specifications.

Note that almost all of the nonconforming shafts are too large, because the process mean is located very near to the upper specification limit. Suppose that we can **recenter** the manufacturing process, perhaps by adjusting the machine, so that the process mean is exactly equal to the nominal value of 0.2500. Then we have

$$\begin{aligned} P\{0.2485 \le x \le 0.2515\} &= P\{x \le 0.2515\} - P\{x \le 0.2485\} \\ &= \Phi\left(\frac{0.2515 - 0.2500}{0.0005}\right) - \Phi\left(\frac{0.2485 - 0.2500}{0.0005}\right) \\ &= \Phi(3.00) - \Phi(-3.00) \\ &= 0.99865 - 0.00135 \\ &= 0.9973 \end{aligned}$$

By recentering the process we have increased the yield of the process to approximately 99.73%.



■ **FIGURE 3.19** Distribution of shaft diameters, Example 3.8.

EXAMPLE 3.9 Another Use of the Standard Normal Table

Sometimes instead of finding the probability associated with a particular value of a normal random variable, we find it necessary to do the opposite—find a particular value of a normal random variable that results in a given probability. For example, suppose that $x \sim N(10, 9)$. Find the value of x—say, a—such that $P\{x > a\} = 0.05$.

SOLUTION

From the problem statement, we have

$$P\{x > a\} = P\left\{z > \frac{a-10}{3}\right\} = 0.05$$

or

$$P\left\{z \le \frac{a-10}{3}\right\} = 0.95$$

From Appendix Table II, we have $P\{z \le 1.645\} = 0.95$, so

$$\frac{a-10}{3} = 1.645$$

or

$$a = 10 + 3(1.645) = 14.935$$

The normal distribution has many useful properties. One of these is relative to **linear combinations** of normally and independently distributed random variables. If $x_1, x_2 \ldots, x_n$ are normally and independently distributed random variables with means $\mu_1, \mu_2, \ldots, \mu_n$ and variances $\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2$, respectively, then the distribution of

$$y = a_1x_1 + a_2x_2 + \cdots + a_nx_n$$

is normal with mean

$$\mu_y = a_1\mu_1 + a_2\mu_2 + \cdots + a_n\mu_n \tag{3.27}$$

and variance

$$\sigma_y^2 = a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \cdots + a_n^2\sigma_n^2 \tag{3.28}$$

where $a_1, a_2, \ldots, a_n$ are constants.

The Central Limit Theorem. The normal distribution is often assumed as the appropriate probability model for a random variable. Later on, we will discuss how to check the validity of this assumption; however, the central limit theorem is often a justification of approximate normality.

Definition

The Central Limit Theorem if $x_1, x_2, \ldots, x_n$ are independent random variables with mean μ_i and variance σ_i^2, and if $y = x_1 + x_2 + \cdots + x_n$, then the distribution of

$$\frac{y - \sum_{i=1}^{n} \mu_i}{\sqrt{\sum_{i=1}^{n} \sigma_i^2}}$$

approaches the $N(0, 1)$ distribution as n approaches infinity.

The central limit theorem implies that the sum of n independently distributed random variables is approximately normal, regardless of the distributions of the individual variables. The approximation improves as n increases. In many cases the approximation will be good for small n—say, $n < 10$—whereas in some cases we may require very large n—say, $n > 100$—for the approximation to be satisfactory. In general, if the x_i are identically distributed, and the distribution of each x_i does not depart radically from the normal, then the central limit theorem works quite well for $n \geq 3$ or 4. These conditions are met frequently in quality-engineering problems.

3.3.2 The Lognormal Distribution

Variables in a system sometimes follow an exponential relationship, say $x = \exp(w)$. If the exponent is a random variable, say w, $x = \exp(w)$ is a random variable and the distribution of x is of interest. An important special case occurs when w has a normal distribution. In that case, the distribution of x is called a **lognormal distribution.** The name follows from the transformation $\ln(x) = w$. That is, the natural logarithm of x is normally distributed.

Probabilities for x are obtained from the transformation to w, but we need to recognize that the range of x is $(0, \infty)$. Suppose that w is normally distributed with mean θ and variance ω^2; then the cumulative distribution function for x is

$$F(a) = P[x \leq a] = P[\exp(w) \leq a] = P[w \leq \ln(a)]$$
$$= P\left[z \leq \frac{\ln(a) - \theta}{\omega}\right] = \Phi\left[\frac{\ln(a) - \theta}{\omega}\right]$$

for $x > 0$, where z is a standard normal random variable. Therefore, Appendix Table II can be used to determine the probability. Also, $f(x) = 0$, for $x \leq 0$. The lognormal random variable is always nonnegative.

The lognormal distribution is defined as follows.

Definition

Let w have a normal distribution mean θ and variance ω^2; then $x = \exp(w)$ is a **lognormal random variable,** and the lognormal distribution is

$$f(x) = \frac{1}{x\omega\sqrt{2\pi}} \exp\left[-\frac{(\ln(x) - \theta)^2}{2\omega^2}\right] \quad 0 < x < \infty \tag{3.29}$$

The mean and variance of x are

$$\mu = e^{\theta + \omega^2/2} \quad \text{and} \quad \sigma^2 = e^{2\theta + \omega^2}\left(e^{\omega^2} - 1\right) \tag{3.30}$$

The parameters of a lognormal distribution are θ and ω^2, but care is needed to interpret that these are the mean and variance of the normal random variable w. The mean and variance of x are the functions of these parameters shown in equation (3.30). Figure 3.20 illustrates lognormal distributions for selected values of the parameters.

The lifetime of a product that degrades over time is often modeled by a lognormal random variable. For example, this is a common distribution for the lifetime of a semiconductor laser. Other continuous distributions can also be used in this type of application. However, because the lognormal distribution is derived from a simple exponential function of a normal random variable, it is easy to understand and easy to evaluate probabilities.

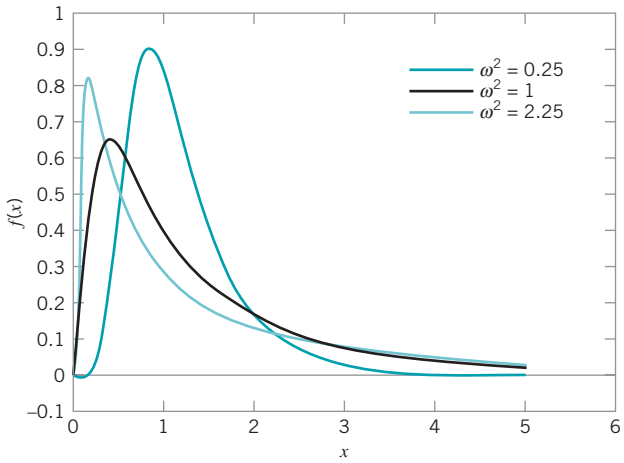


FIGURE 3.20 Lognormal probability density functions with $\theta = 0$ for selected values of ω^2.

EXAMPLE 3.10 Medical Laser Lifetime

The lifetime of a medical laser used in ophthalmic surgery has a lognormal distribution with $\theta = 6$ and $\omega = 1.2$ hours. What is the probability the lifetime exceeds 500 hours?

SOLUTION

From the cumulative distribution function for the lognormal random variable

$$P(x > 500) = 1 - P[\exp(w) \le 500] = 1 - P[w \le \ln(500)]$$

$$= \Phi\left(\frac{\ln(500) - 6}{1.2}\right) = 1 - \Phi(0.1788)$$

$$= 1 - 0.5710 = 0.4290$$

What lifetime is exceeded by 99% of lasers? Now the question is to determine a such that $P(x > a) = 0.99$. Therefore,

$$P(x > a) = P[\exp(w) > a] = P[w > \ln(a)]$$

$$= 1 - \Phi\left(\frac{\ln(a) - 6}{1.2}\right) = 0.99$$

From Appendix Table II, $1 - \Phi(a) = 0.99$ when $a = -2.33$. Therefore,

$$\frac{\ln(a) - 6}{1.2} = -2.33 \quad \text{and} \quad a = \exp(3.204) = 24.63 \text{ hours}$$

Determine the mean and standard deviation of lifetime. Now,

$$\mu = e^{\theta + \omega^2/2} = \exp(6 + 0.72) = 828.82 \text{ hours}$$

$$\sigma^2 = e^{2\theta + \omega^2}\left(e^{\omega^2} - 1\right) = \exp(12 + 1.44)[\exp(1.44) - 1]$$

$$= 2,212,419.85$$

so the standard deviation of the lifetime is 1487.42 hours. Notice that the standard deviation of the lifetime is large relative to the mean.

3.3.3 The Exponential Distribution

The probability distribution of the exponential random variable is defined as follows.

Definition

The **exponential distribution** is

$$f(x) = \lambda e^{-\lambda x} \qquad x \geq 0 \tag{3.31}$$

where $\lambda > 0$ is a constant. The ***mean*** and ***variance*** of the exponential distribution are

$$\mu = \frac{1}{\lambda} \tag{3.32}$$

and

$$\sigma^2 = \frac{1}{\lambda^2} \tag{3.33}$$

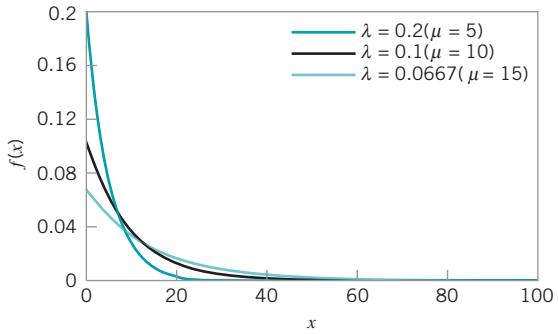
respectively.

Several exponential distributions are shown in Fig. 3.21.

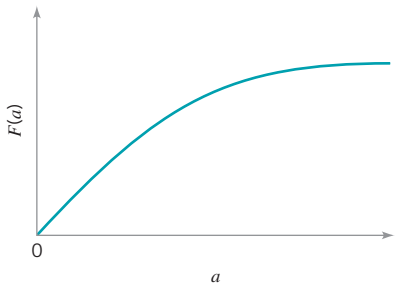
The cumulative exponential distribution is

$$\begin{aligned} F(a) &= P\{x \leq a\} \\ &= \int_0^a \lambda e^{-\lambda t} dt \\ &= 1 - e^{-\lambda a} \qquad a \geq 0 \end{aligned} \tag{3.34}$$

Figure 3.22 illustrates the exponential cumulative distribution function.



■ **FIGURE 3.21** Exponential distributions for selected values of λ.



■ **FIGURE 3.22** The cumulative exponential distribution function.

The exponential distribution is widely used in the field of **reliability engineering** as a model of the time to failure of a component or system. In these applications, the parameter λ is called the **failure rate** of the system, and the mean of the distribution $1/\lambda$ is called the **mean time to failure.**[2] For example, suppose that an electronic component in an airborne radar system has a useful life described by an exponential distribution with failure rate 10^{-4}/h; that is, $\lambda = 10^{-4}$. The mean time to failure for this component is $1/\lambda = 10^4 = 10{,}000$ h. If we wanted to determine the probability that this component would fail before its expected life, we would evaluate

$$P\left\{x \le \frac{1}{\lambda}\right\} = \int_0^{1/\lambda} \lambda e^{-\lambda t}\, dt = 1 - e^{-1} = 0.63212$$

This result holds regardless of the value of λ; that is, the probability that a value of an exponential random variable will be less than its mean is 0.63212. This happens, of course, because the distribution is not symmetric.

There is an important relationship between the exponential and Poisson distributions. If we consider the Poisson distribution as a model of the number of occurrences of some event in the interval $(0, t]$, then from equation 3.15 we have

$$p(x) = \frac{e^{-\lambda t}(\lambda t)^x}{x!}$$

Now $x = 0$ implies that there are no occurrences of the event in $(0, t]$, and $P\{x = 0\} = p(0) = e^{-\lambda t}$. We may think of $p(0)$ as the probability that the interval to the first occurrence is greater than t, or

$$P\{y > t\} = p(0) = e^{-\lambda t}$$

where y is the random variable denoting the interval to the first occurrence. Since

$$F(t) = P\{y \le t\} = 1 - e^{-\lambda t}$$

and using the fact that $f(y) = dF(y)/dy$, we have

$$f(y) = \lambda e^{-\lambda y} \tag{3.35}$$

as the distribution of the interval to the first occurrence. We recognize equation (3.35) as an exponential distribution with parameter λ. Therefore, we see that if the number of occurrences of an event has a Poisson distribution with parameter λ, then the distribution of the interval *between* occurrences is exponential with parameter λ.

3.3.4 The Gamma Distribution

The probability distribution of the gamma random variable is defined as follows.

[2]See the supplemental text material for more information.

Definition

The **gamma distribution** is

$$f(x) = \frac{\lambda}{\Gamma(r)}(\lambda x)^{r-1} e^{-\lambda x} \qquad x \geq 0 \tag{3.36}$$

with **shape parameter** $r > 0$ and **scale parameter** $\lambda > 0$. The **mean** and **variance** of the gamma distribution are

$$\mu = \frac{r}{\lambda} \tag{3.37}$$

and

$$\sigma^2 = \frac{r}{\lambda^2} \tag{3.38}$$

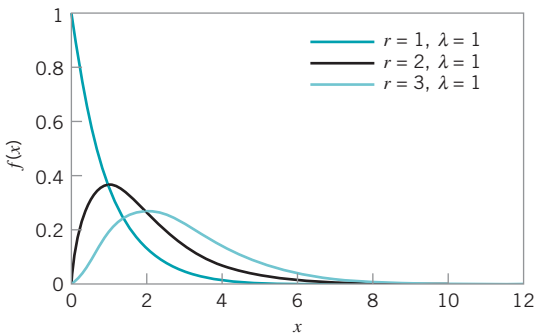
respectively.[3]

Several gamma distributions are shown in Fig. 3.23. Note that if $r = 1$, the gamma distribution reduces to the exponential distribution with parameter λ (Section 3.3.3). The gamma distribution can assume many different shapes, depending on the values chosen for r and λ. This makes it useful as a model for a wide variety of continuous random variables.

If the parameter r is an integer, then the gamma distribution is the sum of r independently and identically distributed exponential distributions, each with parameter λ. That is, if $x_1, x_2, \ldots, x_r$ are exponential with parameter λ and independent, then

$$y = x_1 + x_2 + \cdots + x_r$$

is distributed as gamma with parameters r and λ. There are a number of important applications of this result.

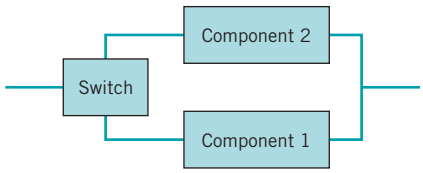


■ **FIGURE 3.23** Gamma distributions for selected values or r and $\lambda = 1$.

[3] $\Gamma(r)$ in the denominator of equation 3.36 is the gamma function, defined as $\Gamma(r) = \int_0^\infty x^{r-1} e^{-x} dx$, $r > 0$. If r is a positive integer, then $\Gamma(r) = (r-1)!$

EXAMPLE 3.11 A Standby Redundant System

Consider the system shown in Fig. 3.24. This is called a **standby redundant system,** because while component 1 is on, component 2 is off, and when component 1 fails, the switch automatically turns component 2 on. If each component has a life described by an exponential distribution with $\lambda = 10^{-4}/\text{h}$, say, then the system life is gamma distributed with parameters $r = 2$ and $\lambda = 10^{-4}$. Thus, the mean time to failure is $\mu = r/\lambda = 2/10^{-4} = 2 \times 10^4$ h.



■ **FIGURE 3.24** The standby redundant system for Example 3.11.

The cumulative gamma distribution is

$$F(a) = 1 - \int_a^\infty \frac{\lambda}{\Gamma(r)}(\lambda t)^{r-1} e^{-\lambda t} dt \tag{3.39}$$

If r is an integer, then equation 3.39 becomes

$$F(a) = 1 - \sum_{k=0}^{r-1} e^{-\lambda a} \frac{(\lambda a)^k}{k!} \tag{3.40}$$

Consequently, the cumulative gamma distribution can be evaluated as the sum of r Poisson terms with parameter λa. This result is not too surprising, if we consider the Poisson distribution as a model of the number of occurrences of an event in a fixed interval, and the gamma distribution as the model of the portion of the interval required to obtain a specific number of occurrences.

3.3.5 The Weibull Distribution

The Weibull distribution is defined as follows.

Definition

The **Weibull distribution** is

$$f(x) = \frac{\beta}{\theta}\left(\frac{x}{\theta}\right)^{\beta-1} \exp\left[-\left(\frac{x}{\theta}\right)^{\beta}\right] \qquad x \geq 0 \tag{3.41}$$

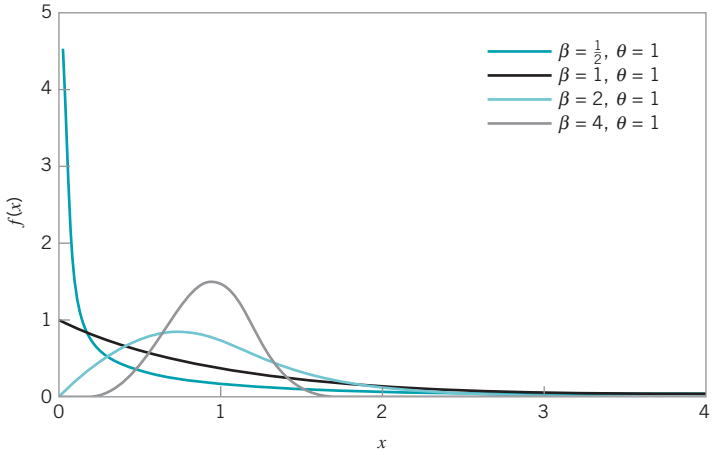
where $\theta > 0$ is the **scale parameter,** and $\beta > 0$ is the **shape parameter.** The **mean** and **variance** of the Weibull distribution are

$$\mu = \theta\Gamma\left(1 + \frac{1}{\beta}\right) \tag{3.42}$$

and

$$\sigma^2 = \theta^2\left[\Gamma\left(1 + \frac{2}{\beta}\right) - \left\{\Gamma\left(1 + \frac{1}{\beta}\right)\right\}^2\right] \tag{3.43}$$

respectively.



■ **FIGURE 3.25** Weibull distributions for selected values of the shape parameter β and scale parameter $\theta = 1$.

The Weibull distribution is very flexible, and by appropriate selection of the parameters θ and β, the distribution can assume a wide variety of shapes. Several Weibull distributions are shown in Fig. 3.25 for $\theta = 1$ and $\beta = 1/2$, 1, 2, and 4. Note that when $\beta = 1$, the Weibull distribution reduces to the exponential distribution with mean $1/\theta$. The cumulative Weibull distribution is

$$F(a) = 1 - \exp\left[-\left(\frac{a}{\theta}\right)^{\beta}\right] \tag{3.44}$$

The Weibull distribution has been used extensively in reliability engineering as a model of time to failure for electrical and mechanical components and systems. Examples of situations in which the Weibull has been used include electronic devices such as memory elements, mechanical components such as bearings, and structural elements in aircraft and automobiles.[4]

EXAMPLE 3.12 Time to Failure for Electronic Components

The time to failure for an electronic component used in a flat panel display unit is satisfactorily modeled by a Weibull distribution with $\beta = \frac{1}{2}$ and $\theta = 5000$. Find the mean time to failure and the fraction of components that are expected to survive beyond 20,000 hours.

SOLUTION

The mean time to failure is

$$\mu = \theta\Gamma\left(1 + \frac{1}{\beta}\right) = 5000\Gamma\left(1 + \frac{1}{\frac{1}{2}}\right)$$

$$= 5000\Gamma(3) = 10{,}000 \text{ hours}$$

The fraction of components expected to survive $a = 20{,}000$ hours is

$$1 - F(a) = \exp\left[-\left(\frac{a}{\theta}\right)^{\beta}\right]$$

or

$$1 - F(20{,}000) = \exp\left[-\left(\frac{20{,}000}{5{,}000}\right)^{\frac{1}{2}}\right]$$

$$= e^{-2}$$

$$= 0.1353$$

That is, all but about 13.53% of the subassemblies will fail by 20,000 hours.

[4]See the supplemental text material for more information.